Queestion 1

Show that $(\alpha T)^* = \alpha^{-1} T^*$

Solution:

$$< x, (\alpha T^*)^* y > = < (\alpha T)x, y >$$
$$< x, (\alpha T^*)^* y > = < \alpha(Tx), y >$$
$$< x, (\alpha T^*)^* y > = \bar{\alpha} < Tx, y >$$
$$< x, (\alpha T^*)^* y > = \bar{\alpha} < x, T^* y >$$
$$< x, (\alpha T^*)^* y > = < x, \bar{\alpha} T^* y >$$
$$(\alpha T^*)^* = \bar{\alpha} T^*$$

Question 2

Prove that $(S+T)^{*} = S^{*} + T^{*}$

Prove:

$$< x, (S+T)^{*}y > = < (S+T)x, y >$$
$$< x, (S+T)^{*}y \geq < (Sx + Tx), y >$$
$$< x, (S+T)^{*}y > = < Sx, y > + < Tx, y >$$
$$< x, (S+T)^{*}y > = < x, S^{*}y > + < x, T^{*}y >$$
$$< x, (S+T)^{*}y > = < x, (S^{*}y + T^{*}y) >$$
$$< x, (S+T)^{*}y > = < x, (S^{*} + T^{*})y >$$

Hence we have proved that

$$(S+T)^{*} = S^{*} + T^{*}$$

Question 3

Find matrix multiplication if $(\xi_1, \xi_2, \xi_3) \leftrightarrow (\xi_1, \xi_2, -\xi_1 - \xi_2)$

Solution:

$$\begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}\begin{bmatrix} \xi_1 \\ \xi_2 \\ \xi_3 \end{bmatrix} = \begin{bmatrix} \xi_1 \\ \xi_2 \\ -\xi_1, -\xi_2 \end{bmatrix}$$

Hence required matrix is

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -1 & -1 & 0 \end{bmatrix}$$

Question 3

Find matrix multiplication if $(\xi_1, \xi_2, \xi_3) \leftrightarrow (\xi_1, \xi_2, -\xi_1 - \xi_2)$

Solution:

$$\begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}\begin{bmatrix} \xi_1 \\ \xi_2 \\ \xi_3 \end{bmatrix} = \begin{bmatrix} \xi_1 \\ \xi_2 \\ -\xi_1, -\xi_2 \end{bmatrix}$$

Hence required matrix is

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -1 & -1 & 0 \end{bmatrix}$$

Question 4

$$\|x_1 + x_2 + \dots\dots\dots + x_n\|^2 = \|x_1\|^2 + \|x_2\|^2 + \dots\dots + \|x_n\|^2$$

Prove:

$$\left\|\sum_{i=1}^{n} x_i\right\|^2 = \left\langle \left\|\sum_{i=1}^{n} x_i, \sum_{i=1}^{n} x_i\right\| \right\rangle$$

$$= < x_1 + \dots\dots\dots + x_n, x_1 + \dots\dots\dots + x_n >$$

$$= < x_1, x_1 + \dots\dots\dots + x_n > + \dots\dots + < x_n, x_1 + \dots\dots\dots + x_n >$$

$$= \sum_{i=1}^{n}\sum_{j=1}^{n} < x_i, x_j >$$

$< x_i, x_j > = \|x_i\|^2$, if $i \neq j$ $\quad < x_i, x_j > = 0$ and for $i = j$ then $< x_i, x_j > = \|x_i\|^2$

$$\left\|\sum_{i=1}^{n} x_i\right\|^2 = \sum_{i=1}^{n} \|x_i\|^2$$

Question 5

Prove that $(ST)^* = S^*T^*$

Prove:

$$< x,(ST)^*y >=< (ST)x,y >$$
$$< x,(ST)^*y \geq< (SxTx),y >$$
$$< x,(ST)^*y >=< Sx,y >< Tx,y >$$
$$< x,(ST)^*y >=< x,S^*y >< x,T^*y >$$
$$< x,(ST)^*y >=< x,(S^*+T^*y) >$$
$$< x,(ST)^*y >=< x,(S^*T^*)y >$$

Hence we have proved that

$$(ST)^* = S^*T^*$$

Question 6

Show that $\overline{N(T)} < N(T)$

Prove:

$$x \in N(T)$$
$$\exists\ (x_n) \textit{such that}\ x_n \ \in N(T) + x_n \to x$$

Therefore

$$Tx_n \to Tx$$
$$x_n \ \in N(T)$$
$$Tx_n = 0$$
$$T_n = 0$$
$$x \ \in N(T)$$
$$\overline{N(T)} < N(T)$$
$$\overline{N(T)} = N(T)$$
$$N(T) \textit{is closed.}$$

Question 7

Prove that $< x, ay + bz > = \bar{\boldsymbol{a}} < x, y > + \bar{\boldsymbol{b}} < x, z >$

Prove:

$$< x, ay + bz > = < \overline{\mathrm{ay} + \mathrm{bz}, \mathrm{x}} >$$
$$< x, ay + bz > = < \overline{\mathrm{ay}, \mathrm{x}} > + < \overline{\mathrm{bz}, \mathrm{x}} >$$
$$< x, ay + bz > = \bar{\mathrm{a}} < \overline{\mathrm{y}, \mathrm{x}} > + \bar{\mathrm{b}} < \overline{\mathrm{z}, \mathrm{x}} >$$
$$< x, ay + bz > = \bar{\mathrm{a}} < x, y > + \bar{\mathrm{b}} < x, z >$$

Hence proved.

Question 8

Let H_1 and H_2 be Hilbert spaces, $S: H_1 \to H_2$ and $T: H_1 \to H_2$ be bounded linear operator and α any scalar. Then show that $(T^*)^* = T$.

Prove:

$$< x,(T^*)^* y \geq < xT^*, y >$$

$$< x,(T^*)^* y \geq < x, Ty >$$

It implies that

$(T^*)^* = T$ so its proved.

Question 9

Prove that the product of two bounded self-adjoint linear operators S and T on a Hilbert space H is self-adjoint if and only if the operators commute, $ST = TS$.

Prove:

$$< x,(ST)^{*}y >=< (ST)x, y >$$
$$< x,(ST)^{*}y \geq < (SxTx), y >$$
$$< x,(ST)^{*}y >=< Sx, y >< Tx, y >$$
$$< x,(ST)^{*}y >=< x, S^{*}y >< x, T^{*}y >$$
$$< x,(ST)^{*}y >=< x,(S^{*}+T^{*}y) >$$
$$< x,(ST)^{*}y >=< x,(S^{*}T^{*})y >$$

Hence we have proved that

$$(ST)^{*} = S^{*}T^{*}$$

$$(ST)^{*} = T^{*}S^{*} = TS.$$

Hence

$$ST = (ST)^{*} \quad \Leftrightarrow \quad ST = TS.$$

Question 10

Let H_1 and H_2 be Hilbert spaces, $S: H_1 \to H_2$ and $T: H_1 \to H_2$ be bounded linear operator and α any scalar. Then show that $\|T^*T\| = \|TT^*\| = \|T\|^2$.

Prove:

$$\|Tx\|^2 = \langle Tx, Tx\rangle = \langle T^*Tx, x\rangle \leq \|T^*Tx\| \, \|x\| \leq \|T^*T\| \, \|x\|^2.$$

Taking the supremum over all x of norm 1, we obtain $\|T\|^2 \leq \|T^*T\|$. we have

$$\|T\|^2 \leq \|T^*T\| \leq \|T^*\| \, \|T\| = \|T\|^2.$$

Hence $\|T^*T\| = \|T\|^2$. Replacing T by T^* and we also have

$$\|T^{**}T^*\| = \|T^*\|^2 = \|T\|^2.$$

Here $T^{**} = T$

Question 11

Let the operators $U : H \to H$ and $V : H \to H$ be unitary; here, H is a Hilbert space. Then show that U is isometric; thus $\|Ux\| = \|x\|$ for all $x \in H$.

Prove:

$$\| U_x \|^2 = < U_x, U_x >$$
$$\| U_x \|^2 = < x, U^* U_x >$$
$$\| U_x \|^2 = < x, Ix >$$
$$\| U_x \|^2 = < x, x >$$
$$\| U_x \|^2 = \| x \|^2$$

It proved.

Question 12

Show that $< T^{*}y, x > = < y, Tx >$

Prove:

$$< T^{*}y, x > = < \overline{y, T^{*}x} >$$
$$< T^{*}y, x > = < \overline{Tx, y} >$$
$$< T^{*}y, x > = < y, Tx >$$

Hence its proved.

Question 13

Let X and Y be inner product spaces and $Q:X \to Y$ a bounded linear operator. Then Show that $Q=0$ if and only if $\langle Qx, y\rangle = 0$ for all $x \in X$ and $y \in Y$.

Prove:

$$Q = 0 \Longrightarrow Qx = 0 \; for \; all \; x\epsilon X$$
$$< Qx, y > = < 0, y >$$
$$< Qx, y > = < 0.w, y >$$
$$< Qx, y > = \; 0 < w, y >$$
$$< Qx, y > = \; 0. \quad for \; all \; x\epsilon X$$

It implies that Qx=0 and by definition Q=0. Its proved.

Question 14

Let the operators $U : H \to H$ and $V : H \to H$ be unitary; here, H is a Hilbert space. Then show that $\|U\| = 1$, provided $H \neq \{0\}$.

OR

Let the operators $U : H \to H$ and $V : H \to H$ be unitary; here, H is a Hilbert space. Then show that $U^{-1} (= U^*)$ is unitary.

Prove:

Since U is bijective it shows that U^{-1} is bijective

$$(U^{-1})^* = (U^*)^* = U = (U^{-1})^{-1}.$$

Hence it implies that U^{-1} is unitary.

Question 15

Let the operators $U : H \to H$ and $V : H \to H$ be unitary; here, H is a Hilbert space. Then show that U is normal.

Prove:

$$UU^* = U^*U$$

Since we know that

$$U^* = U^{-1}$$

So

$$UU^{-1} = U^{-1}U$$

$(UV)^* = (UV)^{-1} = I$ Hence its proved that U is normal.

Question 16

In an inner product space V and x, y in V if $x \perp y$ then

$$\|x+y\|^2 = \|x\|^2 + \|y\|^2$$

Prove:

$$\|x+y\|^2 = \langle x+y, x+y \rangle$$

$$= \langle x,x \rangle + \langle x,y \rangle + \langle y,x \rangle + \langle y,y \rangle$$

As x and y are perpendicular so $\langle x,y \rangle = 0, \langle y,x \rangle = 0$

$$\|x+y\|^2 = \langle x,x \rangle + \langle y,y \rangle = \|x\|^2 + \|y\|^2$$

Question 17

$$\|x+y\|^2 + \|x-y\|^2 = 2\left(\|x\|^2 + \|y\|^2\right) \quad \text{for all } x,y \in V$$

Prove:

$$\|x+y\|^2 = \langle x+y, x+y \rangle$$

$$= \langle x,x \rangle + \langle x,y \rangle + \overline{\langle x,y \rangle} + \langle y,y \rangle$$

$$= \|x\|^2 + 2\,\mathrm{Re}\langle x,y \rangle + \|y\|^2 \qquad \text{....(i}$$

Replace y=-y

$$\|x-y\|^2 = \langle x+y, x+y \rangle$$

$$= \langle x,x \rangle - \langle x,y \rangle - \overline{\langle x,y \rangle} + \langle y,y \rangle$$

$$= \|x\|^2 - 2\,\mathrm{Re}\langle x,y \rangle + \|y\|^2 \qquad \text{.....(ii}$$

Adding (i and (ii

$$\|x+y\|^2 + \|x-y\|^2 = 2\|x\|^2 + 2\|y\|^2$$

That we have to prove.

Question18

For any two elements x,y is an inner product space V,

$$|<x,y>| \leq \|x\|.\|y\|$$

Prove:

If x=y=0 then 0=0

Let at least one of x and y is not equal to zero

Let $\quad |<x+\lambda y, x+\lambda y>| \geq 0 \quad$ by definition

$$<x, x+\lambda y> + <\lambda y, x+\lambda y>$$

$$<x, x+\lambda y> + y<y, x+\lambda y>$$

Question 19

SHOW THAT T is uniquely determinal if the image $y_k = Te_k$ of n basis vectors $e_1, \ldots\ldots\ldots, e_n$ are prescribed.

Prove:

$$y = Tx \quad ; y \in Y \quad \{b_1, \ldots\ldots\ldots, b_r\}$$

$$y = \eta_1 b_1 + \eta_2 b_2 + \ldots\ldots\ldots + \eta_r b_r$$

$$Te_k \in Y, \qquad Te_1 = \tau_{11} b_1 + \tau_{12} b_2 + \ldots\ldots\ldots + \tau_{1r} b_r$$

$$Te_k = \sum_{j=1}^{r} \tau_{kj} b_j$$

$$y = \sum_{j=1}^{r} \eta_j b_j = \sum_{k=1}^{n} \xi_k Te_k = \sum_{k=1}^{n} \xi_k \sum_{j=1}^{r} \tau_{kj} b_j$$

Comibinig these two summation

$$y = \sum_{j=1}^{r} \left(\sum_{k=1}^{n} \tau_{kj} \xi_k \right) b_j$$

$$\eta_j = \sum_{k=1}^{n} \tau_{kj} \xi_k$$

The image $y = Tx = \sum \eta_j b_j$ of $x = \sum \xi_k Te_k$ can be obtained from

$$\eta_j = \sum_{k=1}^{n} \tau_{kj} \xi_k$$

Question 20

Sum and Scalar multipication of Dual.

Prove:

as follows. The SUM $f_1 + f_2$ of two functionals f_1 and f_2 is the functional s whose value at every $x \in X$ is

$$s(x) = (f_1 + f_2)(x) = f_1(x) + f_2(x);$$

the Product αf of a scalar α and a functional f is the functional p whose value at $x \in X$ is

$$p(x) = (\alpha f)(x) = \alpha f(x).$$

Question 21

Define Inner Product Space:

The pair $(V, <\bullet,\bullet>)$ is called an inner product space.

Define Appolonius Identity:

$$\|z-x\|^2+\|z-y\|^2=\frac{1}{2}\|x-y\|^2+2\left\|z-\frac{1}{2}(x+y)\right\|^2 \quad , \quad x,y,z\in V$$

Define Equal Operators:

Let T_1 and T_2 be two operators defined on equal i.e. $T_1 = T_2$ if they have the same domain i.e. $D(T_1) = D(T_2)$

Question 22

Define Orthonormal System.

Orthonormal systems are those functions which are real or complex whose inner product with itself results in 1 and with other functions result in 0.

Define Polarization Identity.

For any x, y in complex inner product space

$$<x, y> = \frac{1}{4}\left\{\|x+y\|^2 - \|x-y\|^2 + i\|x+iy\|^2 - i\|x-iy\|^2\right\}$$

We have to prove this complex inner product space.

Question 23

Define Bounded Linear Function.

Let X and Y be normed spaces and $T : D(T) \to Y$ a linear operator, where $D(T) \subset X$. The operator T is said to be bounded if there is a real number c such that for all $x \in D(T)$.

$\|Tx\| \le c\|x\|$

Define Canonical Mapping.

$C : X \to X^{**}$ this mapping is called canonical mapping of X into X^{**} defined as $x \mapsto g_x$.

Question 24

Define Unitary Operator.

Abounded Linear Operator $T: H \to H$ on a Hilbert Space H is said to be Unitary Operator if T is bijective and $T^* = T^{-1}$.

Define Normal Operator.

Abounded Linear Operator $T: H \to H$ on a Hilbert Space H is said to be Unitary Operator if T is bijective and $TT^* = T^*T$

Define Extension of T.

Extension of T on a set $M \supset D(T)$ is defined as $\tilde{T}: M \to Y$ or

$$\tilde{T}x = Tx \; for \; all \; x \in D(T)$$

Question 25

Write examples of Bounded Linear Operators

Identity Operator, Zero Operator, Differentiation Operator, Integral Operator

Define Self Adjoint Operator or Hermitian Operator.

Abounded Linear Operator $T: H \to H$ on a Hilbert Space H is said to be Self Adjoint Operator or Hermitian Operator if $T^* = T$.

Give an example of function which is not linear.

$\|\,.\,\|: x \to R$ is functional but not linear.

Find basis of R(T).

$(\xi_1, \xi_2, -\xi_1, -\xi_2) = \xi_1(1,0,-1) + \xi_2(0,1,-1)$

So the basis of R(T)= $\{(1,0,-1), (0,1,-1)\}$

Question 26

Give any two examples of Linear Functional.

Space C[a, b] and Space l^2

Define Linear Functional:

A linear functional f is a linear operator with domain in a vector space X and range in the scalar field K of X; thus $f: D(f) \rightarrow K$. where K=R if X is real and K=C if X is complex.

Define Bounded Linear Extension:

$T: D(T) \rightarrow Y$ be a bounded linear operator, where D(T) lies in a normal space X and Y is a Banach Space then T has an extension $\tilde{T}: \overline{D(T)} \rightarrow Y$. Where $\tilde{T}$ is a bounded linear operator of norm.

$$\|\tilde{T}\| = \|T\|$$